Bihar Engineering University, Patna

B.Tech 6th Semester Examination, 2024

Course : B.Tech
Code : 100612

Subject: Digital Control Systems

Time: 03 Hours
Full Marks: 70

Instructions:-
(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct answer the following (Any seven question only): [2 x 7 = 14]

(a) What is the primary role of a sample and hold circuit in digital control systems?
(i) To amplify the input signal.
(ii) To convert a continuous-time signal into a discrete-time signal.
(iii) To convert an analog signal into a continuous signal.
(iv) To filter noise from the system.

(b) What is the Z-transform primarily used for in discrete-time systems?
(i) To convert continuous systems into the frequency domain.
(ii) To convert discrete-time signals into the frequency domain.
(iii) To solve differential equations.
(iv) To calculate the Laplace transform.

(c) The ZOH (Zero-Order Hold) equivalent of a system is used to:
(i) Determine the time delay of a continuous system.
(ii) Represent the discrete-time equivalent of a continuous-time system.
(iii) Model the effect of quantization in a discrete-time system.
(iv) Choose the appropriate sampling frequency.

(d) The pulse transfer function of a system is analogous to which of the following in continuous-time systems?
(i) State-space representation
(ii) Transfer function
(iii) Root locus diagram
(iv) Nyquist plot

(e) Which of the following methods can be used to analyze the stability of discrete-time systems?
(i) Nyquist Criterion
(ii) Bode Plot
(iii) Bilinear Transformation
(iv) Routh-Hurwitz Criterion

(f) In discrete-time systems, which property ensures that the system's states can be driven to any desired value through suitable inputs?
(i) Observability
(ii) Stability
(iii) Controllability
(iv) Reachability

(g) The effect of pole-zero cancellation in discrete-time systems typically impacts which of the following properties?
(i) Stability
(ii) Frequency response
(iii) Controllability and observability
(iv) Time response

© Apna BEU

(h) A discrete state feedback controller is designed primarily to:
(i) Stabilize continuous-time systems.
(ii) Modify the system's state variables for desired performance.
(iii) Track the desired output for discrete systems.
(iv) Minimize the effects of sampling frequency.

(i) Fast Output Sampling (FOS) is a technique primarily used to:
(i) Increase the sampling frequency of a continuous system.
(ii) Estimate the state variables of a continuous system.
(iii) Control a system using only output measurements at discrete time intervals.
(iv) Design an optimal continuous-time controller.

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play

(j) Which of the following best describes the periodic output feedback controller design in digital control systems?

(i) It adjusts the controller parameters continuously over time.

(ii) It uses random output samples to control the system.

(iii) It samples system outputs at regular intervals and applies control inputs accordingly.

(iv) It directly controls the system's states without considering the output.

Q.2 (a) Explain the process of obtaining a discrete representation of a continuous-time system. [7]

(b) Discuss the role of the sample and hold circuit and the impact of sampling and quantization on system performance. [7]

Q.3 (a) Define the Z-Transform and explain its significance in the analysis of discrete-time systems. How is the Z-transform used to solve difference equations in control systems? [7]

(b) For a discrete-time system with the difference equation $y[n+2] + 3y[n+1] + 2y[n] = x[n]$, determine the pulse transfer function using the Z-transform. Solve for the output response if the input is a unit step. [7]

Q.4 (a) What is the Jury stability test, and how is it used to determine the stability of discrete-time systems? [7]

(b) Apply the Jury stability criterion to analyze the stability of a discrete-time system with the characteristic equation $z^3 + 0.5z^2 - 0.3z - 0.2 = 0$. Determine if the system is stable and show all steps. [7]

Q.5 (a) Explain the concept of controllability and observability in the state-space analysis of discrete-time systems. Discuss how pole-zero cancellation affects these properties. [7]

(b) Consider a discrete-time system with the state-space model: [7]

$$x[k+1] = \begin{bmatrix} 0.5 & 1 \\ 0 & 0.5 \end{bmatrix} x[k] + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u[k]$$

$$y[k] = [1 \quad 0]\, x[k]$$

© Apna BEU

Test the system for controllability and observability using the relevant matrices.

Q.6 (a) Discuss the design of a discrete PID controller. Explain how each component (Proportional, Integral, and Derivative) affects the performance of a digital control system. [7]

(b) Describe the key steps in designing a set point tracker for a discrete-time system. How does the design ensure that the system output accurately follows the desired reference signal? [7]

Q.7 (a) Describe the concept of Fast Output Sampling (FOS) in digital control systems. How does it differ from traditional output feedback techniques? [7]

(b) Design a periodic output feedback controller for a discrete-time system. Use the following state-space model: [7]

$$x[k+1] = \begin{bmatrix} 0.8 & 0.1 \\ 0 & 0.9 \end{bmatrix} x[k] + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u[k]$$

$$y[k] = [1 \quad 0]\, x[k]$$

Demonstrate the steps involved in designing the feedback controller.

Q.8 (a) Why is the choice of sampling frequency crucial in digital control systems, and what issues can arise with inappropriate sampling rates? [7]

(b) What is the significance of the pulse transfer function in the analysis of discrete-time control systems? Explain how it is derived and its role in system stability and time response. [7]

Q.9 (a) What is Lyapunov stability in the context of discrete-time systems? Discuss how the Lyapunov function is used to determine the stability of a system in state-space representation. [7]

(b) Compare the advantages and limitations of the Jury test and bilinear transformation for assessing the stability of discrete-time systems. In which scenarios would one method be preferred over the other? [7]

Apna BEU: Syllabus, PYQs & AI
All-in-One app for Syllabus, PYQs, Result, Attendance and more.
GET IT ON Google Play